

PA 528: Program Evaluation

Fall 2026

Time and Location: Monday, 6:00 pm – 8:45 pm, 2232 AEH

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Instructor Office Hours: TBA (but I'm happy to meet any time)

Teaching Assistant: Marietta Lopez Moreira

Teaching Assistant Office Hours: TBA. Email to set up other times.

COURSE OVERVIEW AND OBJECTIVES

A core tenet of good governance is making decisions based on evidence. In this class, we will develop a set of strategies and tools to produce evidence. This evidence will help us understand 'what works' in terms of the policies and programs that various government agencies/departments, nonprofits, schools, and international NGOs implement. Our focus will be on identifying a causal relationship between a policy or program of interest and the outcomes it is designed to achieve. While there are many activities that fall under the umbrella of program evaluation, our work will emphasize impact evaluations. Ultimately, we want to be able to tell funders, political leaders, nonprofit directors, and other stakeholders that a particular program produced or failed to produce its intended results. Such work is critical for enhancing accountability, producing efficient allocation of taxpayer dollars, and delivering services to citizens that improve their lives.

Rossi (2003) stated: "The findings of the majority of evaluations purporting to be impact assessments are not credible. They are not credible because they are built upon research designs that cannot be safely used for impact assessments." And this statement remains true today. In this course, we will explore a range of credible designs for making causal claims from data. These designs include traditional experimental designs, but also quasi-experimental designs, and approaches for making causal inferences from observational data. This course will develop skills in data science, statistics, and causal thinking. We will also address ethical issues and data limits for testing public programs. By the end of the course you will be able to:

- Identify different types of evaluations and when and why they are used
- Understand the nature of causation
- Diagram logic models to visualize a program's theory of change
- Use directed acyclic graphs (DAGs) to model the data generating process and identify confounders, colliders, and mediators
- Identify different threats to validity that may undermine evaluation findings
- Run and interpret statistical models
- Understand the theory and mechanics of key research designs/strategies, including regression discontinuity, instrumental variables, difference-in-differences, and experiments.
- Consider the ethical implications of evaluation choices

COURSE MATERIALS

Books:

There is one primary textbook for this course:

- Paul J. Gertler et al., *Impact Evaluation in Practice*, 2nd ed. (Inter-American Development Bank; World Bank, 2016). This is a free textbook that can be downloaded here: <https://openknowledge.worldbank.org/handle/10986/25030>

In addition, we will draw on select readings from the following text.

- Nick Huntington-Klein, *The Effect: An Introduction to Research Design and Causality*, 2nd ed. (CRC Press, 2025). You can access a free version of the entire book here: <https://theeffectbook.net/>

You do not need to purchase any texts for this class. The Gertler et al. and Huntington-Klein book are freely available online. In any given week there may be a number of articles, book chapters, podcasts, or videos associated with the lecture. You will find all of these on Canvas.

Software:

We will use R, R Studio, and Quarto in this course to conduct our analyses, manage projects, and produce reports. If you are new to R, or nervous about using the software, don't worry. We will have plenty of time to practice and there are amazing online resources to help along the way. I've been teaching classes in R for well over a decade now, and am certain that with time, practice, and a little googling, you will become proficient in the software.

Your personal R goal in the first few weeks of the course is to become comfortable with data manipulations using the Tidyverse (select, filter, mutate, summarize, group by), data import/export, and basic visualizations. If you are new to R or want to learn more, here are a few good places to start. As always, reach out to me if you need any specific guidance.

- *Posit Recipes* cover basics, visualization, data wrangling, and other topics. You can access the recipes here: <https://posit.cloud/learn/recipes>
- *R for Data Science* 2e by Hadley Wickham. You can access a free copy here: <https://r4ds.hadley.nz/>
- As a UIC student you also have access to *LinkedIn Learning videos*. There are a number of helpful tutorials there ranging from beginner to advanced. <https://web.uillinois.edu/linkedinlearning>

COURSE SCHEDULE

Date	Topics and Readings	Due
Session 1 – August 24	What is program evaluation? Types of evaluations. Social problems and program needs. Introduction to R Projects and Quarto. <i>Readings:</i> -Gertler et al. chapter 1	

	-Rossi et al. chapter 2	
Session 2 – August 31	Program theory and design. Causal inference and potential outcomes. <i>Readings:</i> -Gertler et al. chapters 2 and 3 -Rossi et al. chapter 3 -Huntington-Klein chapters 5	
September 7 – No Class	Labor Day Holiday	
Session 3 – September 14	Regression review, interactions, and confounders. Mapping the data-generating process with DAGs. <i>Readings:</i> -Huntington-Klein chapters 6-10 -Review any previous course's material on regression. For additional resources see chapters 5 and 6 in this online textbook: Statistical Inference via Data Science (moderndive.com)	Assignment 1
Session 4 – September 21	DAGs cont. Randomization and experiments <i>Readings:</i> -Gertler et al. chapter 4 -Ritchie Blog Post: Everything you need to know about breastfeeding and intelligence (sciencefictions.org)	Problem set 1
Session 5 – September 28	Matching and weighting <i>Readings:</i> -Gertler et al. chapter 8 -Heiss, 2020, Causal Inference	Problem set 2
October 5	Exam 1	
Session 6 – October 12 <i>Sync Online Session</i>	Qualitative approaches to evaluation – guest lecture Dr. Kate Albrecht. <i>Readings:</i> -McDavid et al., 2019, Program Evaluation and Performance Measurement, chapter 5: Applying Qualitative Evaluation Methods	
Session 7 – October 19	Regression discontinuity designs <i>Readings:</i> -Gertler et al. chapter 6 -skim Huntington-Klein chapter 20	Problem set 3
Session 8 – October 26	Difference-in-Differences Methods <i>Readings:</i> -Gertler et al. chapter 7 -skim Huntington-Klein chapter 18	Problem set 4

Session 9 – November 2	Instrumental Variables <i>Readings:</i> -Gertler et al. chapter 5 -skim Huntington-Klein chapter 19	Problem set 5
Session 10 – November 9 <i>Async Online Session</i>	Network science in evaluations <i>Readings:</i> -Borgatti et al., 2009, Network Analysis in the Social Sciences -Mathematica, 2009, Making Connections: Using Social Network Analysis for Program Evaluation -Valente et al., 2015, Social Network Analysis for Program Implementation	
Session 11 – November 16	Network science in evaluations cont. Threats to validity. <i>Readings:</i> -Gertler et al. chapter 9 -Huntington-Klein chapter 11 -skim Siciliano et al. Networks in the Public Sector, chapter 1	Assignment 2
Session 12 – November 23 <i>Sync Online Session</i>	Ethics of evaluation; Planning an evaluation; choosing the right methods, measures, and sample. Review for Exam 2. <i>Readings:</i> -Gertler et al. chapters 11-13; skim 15 and 16	Problem set 6
November 30	Exam 2	
December 7	Finals Week	Final Project

COURSE GRADING AND ASSIGNMENTS

Graded work will fall into four categories: class participation; assignments and problem sets; an evaluation project; and exams. Below is just a quick summary of each of these categories. See Canvas for details.

Participation (10%): This is pretty simple. Come to class prepared, engage in the discussions, and complete the in-class exercises. I expect everyone to get 100% here.

Assignments and Problem Sets (25%):

Assignments: There will be two written assignments. Each assignment is connected to the larger evaluation project you will turn in at the end of the semester.

Problem sets: There will be five or six short problem sets. These problem sets provide an opportunity for you to practice the methods and tools we are learning in class.

Evaluation Project (25%): For the evaluation project, you will identify a program of interest and develop an evaluation. You will not need to collect any data for this assignment. Rather, we will use R to simulate data that looks like the data you would collect in a real evaluation. The project itself will include

information on the program, including its activities, impact theory, and outcome measures. You will also develop a causal diagram, identify a suitable research design to identify the causal effect of the program and run the proposed analysis on your simulated data. You will work in groups of 2 or 3 for this project.

Exams (40%): Two exams, a midterm and a final, will be given in class.

COURSE AND UNIVERSITY POLICIES

Academic Integrity: As an academic community, UIC is committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. All members of the campus community—students, staff, faculty, administrators—share the responsibility of insuring that these standards are upheld so that such an environment exists. Instances of academic misconduct by students shall be handled pursuant to the Student Disciplinary Policy. The Student Disciplinary Policy is available online at <https://dos.uic.edu/wp-content/uploads/sites/262/2018/10/DOS-Student-Disciplinary-Policy-2018-2019-FINAL.pdf>.

Special Needs: UIC and the Department of Public Policy, Management, and Analytics, are committed to maintaining a barrier-free environment so individuals with disabilities can fully access programs, services and all activities on campus. The Office of Disability Services works to ensure the accessibility of UIC programs, classes, and services to students with disabilities. Services are available for students who have documented disabilities, including vision or hearing impairments and emotional or physical disabilities. Students with disability/access needs or questions may contact the Office of Disability Services at (312) 413-2183 (voice) or (312) 413-0123 (TTY only). Please feel free to contact me if you need any special accommodations.

Diversity and Inclusion: It is my goal that people from diverse backgrounds and perspectives be included in and served by this course, the Department of Public Policy, Management, and Analytics, and the University of Illinois Chicago. I believe the diversity students bring to this class is a resource that can improve student learning and perspectives. Course materials and activities are designed to be respectful of all types of diversity: gender identity, sexuality, disability, age, socioeconomic status, ethnicity, race, nationality, religion, values, and culture. It is my intent that this class be an environment where all students feel safe expressing their perspectives and opinions.

Campus Advocacy Network: Under the Title IX law you have the right to an education that is free from any form of gender-based violence and discrimination. Crimes of sexual assault, domestic violence, sexual harassment, and stalking are against the law and can be prevented. For more information or for confidential victim-services and advocacy contact UIC's Campus Advocacy Network at 312-413-1025 or visit <http://can.uic.edu/>. To make a report to UIC's Title IX office, contact Rebecca Gordon, EdD at TitleIX@uic.edu or (312) 996-5657.

EVALUATION RELATED PODCASTS and WEBSITES

There are a number of excellent podcasts that focus on evidence-based policy and causal inference. These are great ways to hear about new and interesting research but also to understand how the

researcher went about designing their study and threats to validity they considered. Some podcasts that I recommend are:

- Mathematica's On the Evidence
- Casual Inference
- Evidence First, MRDC
- JPAM's Closer Look
- Probable Causation

A few websites that you may want to check out:

- American Evaluation Association: www.eval.org
- CDC Program Performance and Evaluation: www.cdc.gov/evaluation
- US Dept. of Education Standards, Assessments, and Accountability: www.ed.gov/admins/lead/account/saa.html
- US Government Performance Website: <https://www.performance.gov/> also see equity action plans for different departments here: <https://www.performance.gov/equity>
- US Government Evaluation Website: <https://www.evaluation.gov/> Also see agency-specific evaluation plans here: <https://www.evaluation.gov/2024-04-03-agencies-publish-anual-evaluation-plans/> As well as a learning agenda dashboard: <https://www.evaluation.gov/learning-agenda-questions-dashboard/>
- The Urban Institute's Outcomes Indicator Project: <https://www.urban.org/policy-centers/cross-center-initiatives/performance-management-measurement/projects/nonprofit-organizations/projects-focused-nonprofit-organizations/outcome-indicators-project>
- The Campbell Collaboration: <https://www.campbellcollaboration.org/>